Cenwei Zhang

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Educations

The Chinese University of Hong Kong | Information Engineering *Postgraduate* Sep. 2026 (Expected)

Shanghai Jiao Tong University | BME (EE Track) *Undergraduate* Sep. 2022 — Jun. 2026 (Expected)

GPA: 3.63/4.3. Awarded Academic Scholarships multiple times; Prize in the National Undergraduate Biomedical Engineering Innovation and Entrepreneurship Competition.

Undergrad Research: **Thesis: Geometry-Aware & Distribution-Preserving Pruning for Medical SAM**; Multimodal medical imaging (detection/segmentation/tracking).

Skills

- **Languages&Workflow**: Python, C++; proficient in the PyTorch deep learning framework. Linux, Shell, Git, GitHub, Docker.
- **Technical Foundation**: Solid understanding of **Generative Modeling** theories (Flow Matching, Diffusion, AR) and on-manifold optimal transport mechanisms; proficient in backbone architectures including **Transformer, DiT, and ViT**; familiar with foundational LLM frameworks (via nanoGPT reproduction and fine-tuning); knowledgeable in Embodied AI and VLA models (e.g., pi0, RDT).

Work Experience

DolphinAI Co., Ltd. |*Algorithm Research Intern | Medical Image Generation* Apr. 2025 — Dec. 2025

- Tackled high-frequency noise and semantic decoupling by integrating **structure-aware** strategies into the generative backbone, effectively filtering artifacts while preserving the **Manifold Consistency** of fine-grained pathological features.

Selected Publications

Axiomatic On-Manifold Shapley via Optimal Generative Flows |***Under Review at ICML*** *(CCF-A)*

Cenwei Zhang*, *Lin Zhu**, *Manxi Lin, Lei You*†

- **Theoretical Framework & Error Bounds**: Innovatively proposed a geometric attribution framework based on Flow Matching and Wasserstein-2 Optimal Transport. Rigorously derived and established the gradient line integral as the unique functional satisfying axioms such as reparameterization invariance. Formulated strict mathematical stability bounds, proving that model explanation errors are strictly bounded by the generative flow's approximation error.
- **High-Dimensional Manifold Alignment & Quantitative Superiority**: Proposed a geometric alignment evaluation system comprising Flow Consistency Error (FCE), Structure-Aware Total Variation (SATV), and Geometric Path Straightness (GPS). On high-dimensional image manifolds like CelebA-HQ, drastically reduced FCE by 5 orders of magnitude compared to DDIM and other diffusion baselines (from 2.8×10^5 to 1.78), achieving strict on-manifold interpretations with near-Euclidean spatial efficiency (GPS of 1.01).
- **Visual Fidelity & Structural Consistency**: Empirically validated the non-linear improvement of kinetic energy regularization on attribution stability, boosting feature importance Rank Correlation by 33% (from 0.662 to 0.882). Compared to IG and DDIM, this method completely filters out high-frequency "shattered gradient" noise with a lower SATV (down to 0.003), while comprehensively outperforming existing SOTA paradigms in zero-shot Faithfulness evaluations.

Anatomically-Informed GNN on Ground Reaction Force for Parkinson's Disease Diagnosis |***Accepted to IEEE JBHI*** *(CCF-B)*

Wenhao Li, ***Cenwei Zhang****, Shi Chang, Jingtong Zhao, Guanning Lin*†

- Co-developed the spatiotemporal gait analysis model GraphGaitNet, fusing anatomy-based Graph Convolutional Networks (GCN) and Transformer temporal modeling, achieving a high diagnostic accuracy of **96.8%** for Parkinson's disease.

Projects

LLM Pre-training and Fine-tuning System based on nanoGPT |*Independent Project & Implementation*

Jan. 2026 — Mar. 2026

- Independently implemented the GPT-2 foundation model from scratch using PyTorch, integrating modern architectures like RoPE and the **FlashAttention** acceleration operator. Executed end-to-end pre-training on a single H100 GPU utilizing `bfloat16` mixed precision, achieving a 31% Zero-shot accuracy on the downstream HellaSwag commonsense reasoning task, outperforming the original GPT-2 baseline by approximately 10%.

Personal Summary

- Professional working proficiency in English (IELTS 7.0).
- **Academic & Career Vision**: Incoming Master's student at CUHK. Deeply committed to the frontiers of Embodied AI and MLLMs. Dedicated to **Human-centered Artificial Intelligence**, with a vision to utilize MLLMs as cognitive brains to inject interpretable generative policies into physical robots, closing the interactive loop between digital intelligence and the real physical world.
- **Technical Philosophy**: As a geek-minded researcher, I hold an absolute technical faith in the realization of AGI, firmly believing that true intelligence must evolve towards embodiment to better serve humanity. I maintain a sharp engineering intuition amidst the fast-iterating AI landscape, dedicated to building general-purpose agents capable of perception, reasoning, and execution.
- **Personal Life**: Passionate about city walks, hiking, and amateur photography. I enjoy honing logical deduction and task planning skills through Go (inspired by AlphaGo), and often draw inspiration for constructing digital "world models" from the open-world environment of Genshin Impact.

张岑蔚

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教育经历

香港中文大学 | 计算机科学与技术 (信息工程类)| 硕士研究生 2026.09(预计入学)

上海交通大学 | 生物医学工程 (电子信息类)| 工学学士 2022.09—2026.06(预计)

GPA: 3.63/4.3，获学业奖学金多次，全国大学生生物医学工程创新创业国家级竞赛二等奖。

本科科研：**毕设-面向医学 SAM 部署的分布保持与几何感知分层剪枝框架**；多模态医学影像检测/分割/动态跟踪等。

技术能力

- **语言 & 工作流**: 常用 Python,C++；熟悉深度学习框架 Pytorch。熟悉 Linux, Shell, Git, GitHub，docker。
- **工程底座**: 熟悉 **Generative Modeling** (Flow Matching, Diffusion, AR) 理论及流形最优传输机制；熟悉 **Transformer, DiT, ViT** 骨干网络架构；熟悉基础 LLM 框架 (基于 nanoGPT 复现和微调)；了解 VLA 等 Embodied AI 知识 (pi0，RDT 等)

工作经历

DolphinAI **海豚智声有限公司** | 算法研究实习生 / 图像生成 2025.04—2025.12

- 深入剖析医学影像的数据分布特性，针对高频噪声扰动与细粒度语义解耦难题，在生成式底座中探索 Structure-aware 策略。确保模型在滤除高频伪影的同时，严格保持病理级细粒度特征的流形一致性（Manifold Consistency）。

科研论文

Axiomatic On-Manifold Shapley via Optimal Generative Flows |***CCFA-ICML*** 审稿中

Cenwei Zhang*, *Lin Zhu**, *Manxi Lin, Lei You*†

- **理论框架与误差上界**: 创新性提出基于流匹配与 Wasserstein-2 最优传输的几何归因框架，严格推导并确立了梯度线积分是满足重参数化不变性等公理的唯一泛函。推导出严密的数学稳定性上界，证明模型解释误差被生成流的逼近误差严格控制。
- **高维流形对齐与量化压制**: 提出流一致性误差（FCE）、结构感知全变分（SATV）及路径平直度（GPS）等几何对齐评估体系。在 CelebA-HQ 等高维图像流形上，将 FCE 较 DDIM 等扩散基线惊人地降低了 5 个数量级（2.8×10^5 降至 1.78），并以近乎纯欧几里得的空间效率（GPS 达 1.01）实现严格的在流形（On-manifold）解释。
- **视觉保真度与结构一致性**: 实验证实了动能正则化对归因稳定性的非线性提升，使特征重要性排序相关性（Rank Corr）大幅提升 33%（0.662 升至 0.882）。相比 IG 与 DDIM，本方法以更低的 SATV（降至 0.003）彻底过滤了高频“破碎梯度”噪声，同时在零样本 Faithfulness 保真度评测中全面反超现有 SOTA 范式。

Anatomically-Informed GCN on Ground Reaction Force for Parkinson’s Disease Diagnosis |***CCFB-JBHI*** *Accept*

Wenhao Li, ***Cenwei Zhang****, Shi Chang, Jingtong Zhao, Guanning Lin*†

- 参与研发时空步态分析模型 GraphGaitNet，融合基于解剖学的图卷积网络与 Transformer 时序建模，在帕金森病诊断任务中实现 **96.8** 的高准确率。

项目经历

基于 nanoGPT 的 LLM 模型预训练微调 | 独立项目与实现 2026.01—2026.03

- 独立基于 PyTorch 手写复现 GPT-2 底座，融入 RoPE 及 **FlashAttention** 加速算子，借助 `bfloat16` 混合精度于单张 H100 跑通端到端预训练；在下游 HellaSwag 常识推理任务中实现 31% 的 Zero-shot 准确率，较原版模型提升约 10%。

个人总结

- 可以使用英语进行无障碍工作交流（IELTS 7.0）。
- 即将赴香港中文大学攻读 CS 硕士。未来将持续学习具身智能 Embodied AI & MLLM 的前沿领域。致力于 **Human-centered Artificial Intelligence**，期许以 MLLM 为认知大脑，将可解释的生成式策略注入机器人实体，打通数字智能与真实物理世界的交互闭环。
- 作为一名极客派的研究者，对 AGI 抱有绝对的技术信仰，且坚信真正的智能必然走向服务人类以具身化。习惯在快速迭代的 AI 中保持敏锐的工程直觉，致力于构建能感知、会推理、可执行的通用智能体。
- 热爱 Citywalk & hiking；amateur photographer；喜欢在围棋中的逻辑推演与任务规划 (from AlphaGO)；Genshin Impact 开放世界也是构建灵感的过程。~